POLICY PRATICE AZURE

Azure Policy Lab

Repository with a hands-on lab for creating Azure policies to restrict allowed regions.

Objective

- Create a policy to block resource deployments outside of Europe
- Test creating Storage Accounts and Virtual Machines in regions both outside and inside Europe

Comments

This was my first policy lab in Azure. I felt a bit of difficulty with the JSON part, often running into errors and not understanding why. But after two days, I finally figured out what the problem was, solved it, and continued with the practice.

I really enjoyed the hands-on experience because it allowed me to see how to create a policy and assign it to a scope — in this case, I applied it to a resource group.

I created a policy that blocks the creation of resources outside of West Europe and North Europe, using the `"deny"` effect in the JSON. Since I'm studying for the AZ-104, it was great to reinforce concepts like `deny`, `modify`, `audit`, `mode enforcement`, `allOf`, and `anyOf`.

Then I tried creating a storage account outside of Europe to test the policy, and as expected, the `"deny"` worked and the creation was blocked. I also set up a custom error message to inform the user why the error occurred.

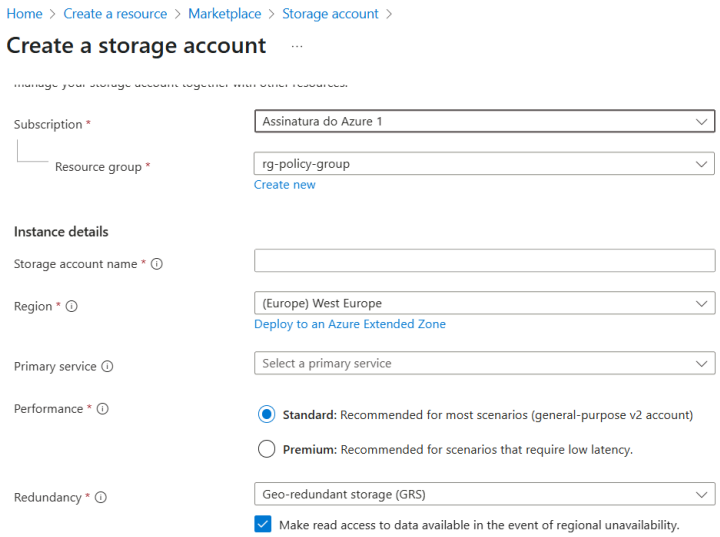
Next, I created a storage account within the allowed region in Europe, and it was successfully created.

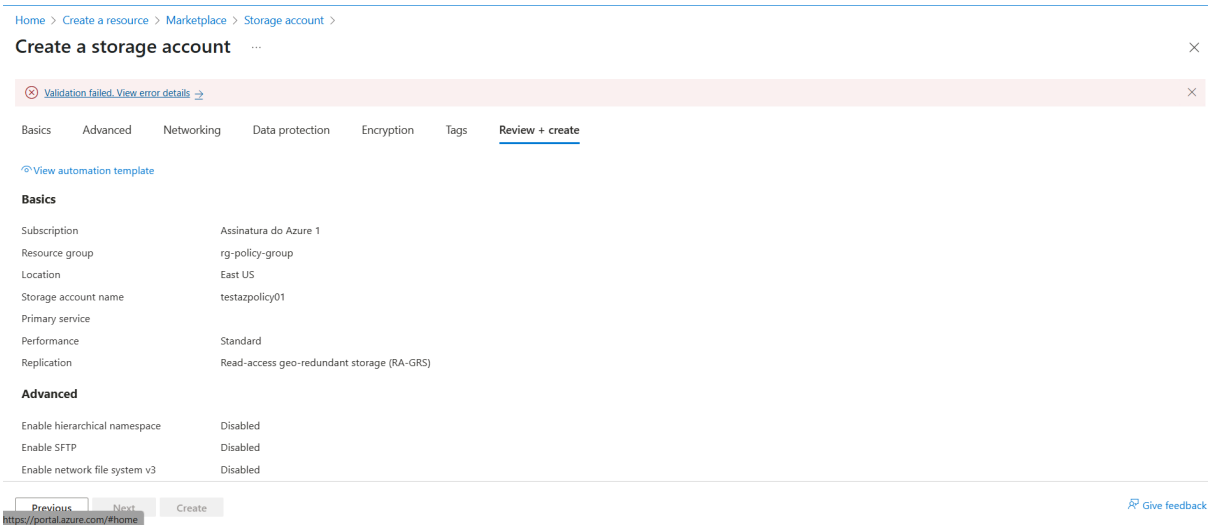
I also tried creating a VM outside of Europe, and it failed as expected, being blocked successfully.

Then I created a VM within West Europe, and it worked, but I came across a detail I didn't know. I had to create a VNet and a subnet because, being my first practice, I hadn't thought of that. So I created the VNet and subnet, and afterward, I saw the policy error blocking the deployment.

Finally, I went into Policy Compliance to check if there were any logs or alerts generated, but there was nothing. That's because when using `"deny"` in the JSON, the block happens right at the resource creation stage, so no logs or alerts were generated.

I was very happy to see the theory turning into practice. On to the next challenges!





Home > Create a resource > Marketplace > Storage account >

Create a storage account

Validation failed. View error details

Basics | Advanced | Networking | Data protection | Encryption | Tags | Review + create

View automation template

Basics

Subscription	Assinatura do Azure 1
Resource group	rg-policy-group
Location	East US
Storage account name	testazpolicy01
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled

Previous | Next | Create

Errors

Summary | Raw Error

ERROR DETAILS

```
Resource 'testazpolicy01' was disallowed by policy. (Code:
RequestDisallowedByPolicy)
```

Policy: Permitir apenas regiões da Europa
Reason: Creating resources outside of Europe is not permitted for compliance and governance reasons.

WAS THIS HELPFUL?

Explain with Copilot

Troubleshooting Options

New Support Request

Give feedback

Tell us about your experience with the ARM Errors page

Home > Create a resource > Marketplace > Virtual machine >

Create a virtual machine

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Assinatura do Azure 1

Resource group * : (New) Resource group

Create new

Instance details

Virtual machine name *

Region * : (Europe) West Europe

Availability options : Availability zone

Zone options :
- (•) Self-selected zone — Choose up to 3 availability zones, one VM per zone
- ( ) Azure-selected zone (Preview) — Let Azure assign the best zone for your needs

Estimated monthly costs

Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.

Give feedback about your estimate experience

Basics	**$87.60**
Virtual machine	$87.60
Image — Ubuntu Server 24.04 LTS	$0.00
Size — Standard_D2s_v3	$87.60
Estimated monthly cost	**$93.41**

< Previous | Next : Disks > | Review + create

Give feedback

Home > Create a resource > Marketplace > Virtual machine >

Create a virtual machine

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me c

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + cr

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. Learn more

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *

Create new

The value must not be empty.

Public IP : (new) vm-test-out-europe-ip

Create new

NIC network security group :
- ( ) None
- (•) Basic
- ( ) Advanced

Public inbound ports * : ( ) None

< Previous | Next : Management > | Review + create

Create virtual network

The Microsoft Azure Virtual Network service enables Azure resources to securely communicate with each other in a virtual network which is a logical isolation of the Azure cloud dedicated to your subscription. You can connect virtual networks to other virtual networks, or your on-premises network. Learn more

Name * : vnet-policy-test

Address space

The virtual network's address space, specified as one or more address prefixes in CIDR notation (e.g. 192.168.1.0/24).

Address range *	Addresses	Overlap
10.0.0.0/16	10.0.0.0 - 10.0.255.255 (65536 addresses)	None
	(0 Addresses)	None

Subnets

The subnet's address range in CIDR notation. It must be contained by the address space of the virtual network.

Subnet name	Address range	Addresses
default	10.0.0.0/24	10.0.0.0 - 10.0.0.255 (256 addresses)
		(0 Addresses)

OK | Discard

Home > Create a resource > Marketplace > Virtual machine >

Create a virtual machine

Validation failed. Click here to view details. →

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | **Review + create**

Price

1 X Standard D2s v3
by Microsoft
Terms of use | Privacy policy

Subscription credits apply
0.1270 USD/hr
Pricing for other VM sizes

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the Azure Marketplace Terms for additional details.

Name: Paulão Austriaco

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Errors

Summary | Raw Error

ERROR DETAILS

```
Resource 'vm-test-out-europe689_z1' was disallowed by policy.
(Code: RequestDisallowedByPolicy)
```

Policy: Permitir apenas regiões da Europa
Reason: Creating resources outside of Europe is not permitted for compliance and governance reasons.

WAS THIS HELPFUL?

Explain with Copilot

Troubleshooting Options

Check Usage + Quota
New Support Request

Give feedback

Tell us about your experience with the ARM Errors page

Home > Create a resource > Marketplace > Virtual machine >

Create a virtual machine

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *: Assinatura do Azure 1
Resource group *: rg-policy-group
Create new

Instance details

Virtual machine name *: vm-test-out-europe
Region *: (Africa) South Africa North
Availability options: Availability zone
Zone options: Self-selected zone
Choose up to 3 availability zones, one VM per zone
Azure-selected zone (Preview)

Estimated monthly costs

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Give feedback about your estimate experience

Basics	**$92.71**
Virtual machine	$92.71
Image Ubuntu Server 24.04 LTS	$0.00
Estimated monthly cost	**$98.42**

< Previous | Next : Disks > | Review + create

Give feedback

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Create a virtual machine

Validation passed

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | **Review + create**

Price

1 X Standard D2s v3
by Microsoft
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Subscription credits apply
0.1200 USD/hr
Pricing for other VM sizes

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Give feedback about your estimate experience

Basics	**$87.60**
Disks	**$1.54**
Networking	**$3.65**
Estimated monthly cost	**$92.79**

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Create a virtual machine

Validation passed

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Price

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by Microsoft
Terms of use | Privacy policy

Subscription credits apply
0.1200 USD/hr
Pricing for other VM sizes

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information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not
provide rights for third-party offerings. See the Azure Marketplace Terms for additional details.

Name: Paulão Austriaco

Generate new key pair

An SSH key pair contains both a public key and a private key. **Azure doesn't store the private key.** After the SSH key resource is created, you won't be able to download the private key again. Learn more

Download private key and create resource

Return to create a virtual machine

Estimated monthly costs

Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.

Give feedback about your estimate experience

Basics	$87.60
Disks	$1.54
Networking	$3.65
Estimated monthly cost	$92.79

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Home >

CreateVm-canonical.ubuntu-24_04-lts-server-20250701221451 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name: CreateVm-canonical.ubuntu-24_04-lts-server-2...
Subscription: Assinatura do Azure 1
Resource group: rg-policy-group

Start time: 7/1/2025, 10:30:08 PM
Correlation ID: 6de6952e-1a52-4190-96c4-dd8e17827f54

Deployment details

Next steps

Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended

Go to resource | Create another VM

Give feedback
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Cost Management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

Microsoft Defender for Cloud
Secure your apps and infrastructure
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Start learning today >

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >

Add or remove favorites by pressing Ctrl+Shift+F

```
{
  "properties": {
    "displayName": "Restringir região para West Europe",
    "policyType": "Custom",
    "mode": "Indexed",
    "description": "Esta política garante que todos os recursos sejam implantados na região West Europe.",
    "metadata": {
      "version": "1.0.0",
      "category": "Location"
    },
    "parameters": {},
    "policyRule": {
      "if": {
        "allOf": [
          {
            "field": "location",
            "notIn": [
              "westeurope"
            ]
          }
        ]
      },
      "then": {
        "effect": "deny"
      }
    }
  }
}
```

Azure services

Create a resource | Policy | Resource groups | Microsoft Entra ID | Budgets | Advisor | Marketplace | Quickstart Center | Azure AI Foundry | More services

Resources

Recent | Favorite

Name	Type	Last Viewed
vm-test-out-europe	Virtual machine	8 minutes ago
rg-policy-group	Resource group	8 minutes ago
testazpolicy01	Storage account	28 minutes ago
Assinatura do Azure 1	Subscription	a week ago

See all

Navigate

Subscriptions | Resource groups | All resources | Dashboard

rtal.azure.com/#

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot | paulo-henrique1967@h... DIRETÓRIO PADRÃO (PAULOHEN...

Home > Policy

Policy | Compliance

Search

- Overview
- Getting started
- Compliance
- Remediation
- Events
- Authoring
  - Definitions
  - Assignments
  - Exemptions

Assign policy | Assign initiative | Refresh

Search

Filter by name or ID... | Scope : Assinatura do Azure 1 | Definition type : All definition types | Compliance state : All compliance states

Overall resource compliance: 100%

Resources by compliance state: 0 (0 - Compliant, 0 - Non-compliant)

Non-compliant initiatives: 0 out of 1

Non-compliant policies: 0 out of 225

Edit columns

Name	Scope	Compliance state	Resource compliance	Non-compliant reso...	Non-compliant poli...
ASC Default (subscription: fc01f5a9-b03	Assinatura do Azure 1	Not started	100% (0 out of 0)	0	0

Give feedback

tps://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Compliance

30°C | POR | 22:39